BT1010 Introduction to Life Sciences



Lecture 2: Biomolecules: Proteins and Enzymes
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Proteins



Types of biomolecules that we eat.



NUTRITIONAL INFORMATION

(TYPICAL VALUES PER 100ml)

Energy	65kcal	Added Sugar	13.3g
Protein	0g	Fat	0g
Carbohydrate	16.2g	Vitamin A	120mcg
Natural Fruit Sugars	2.5g	Calcium	8mg

16.2 gm x 4 = 65 cal



* Nutritional Information per 100g	
1. Energy	562 kcal
2. Carbohydrate	46.10 g
Sugar	0.25 g
3. Total Fat	38.12 g
Saturated Fatty Acids	16.50 g
Monounsaturated Fatty Acids	13.64 g
Polyunsaturated Fatty Acids	3.10 g
Trans Fatty Acids	0 g
4. Protein	8.68 g
5. Fiber	0.83 g
6. Sodium	615 mg
7. Cholesterol	0 mg

* These are approximate values

46.10 gm x 4 = 184.4 cal
8.68 gm x 4 = 34.72 cal
38.12 gm x 9 = 343.08 cal
Total = 562 cal

1 gm of carbohydrates provide 4 calories.
1 gm of proteins provide 4 calories
1 gm of fats/lipids provide 9 calories

Note: Nutritional calorie is the equivalent of the kilocalorie used by chemists and biochemists. So, 65 calorie diet is the equivalent of a 65 kcal diet.

TOP VEGETARIAN PROTEIN SOURCES



Greek Yogurt
23 grams of protein per cup



Lentils
4 grams of protein per 1/4 cup (cooked)



Beans
(chickpeas, black beans, etc.) 4 grams of protein per 1/4 cup



Cottage Cheese
14 grams of protein per 1/2 cup



Hemp Seeds
4 grams of protein per 1 tablespoon



Chia Seeds
3 grams of protein per 1 tablespoon



Edamame
5 grams of protein per 1/4 cup (shelled)



Green Peas
8 grams of protein per cup



Quinoa
8 grams of protein per cup (cooked)



Peanut Butter
3.5 grams of protein per 1 tablespoon



Almonds
3 grams per 1/2 ounce



Eggs
6 grams of protein per large egg

EatingWell

How much protein?



2 eggs (size 6), boiled
13.2g



95g can of tuna, drained
25.1g



100g salmon, pan-fried
20g



100g steak, braised
32.1g



100g chicken breast, grilled
31.2g



1 cup (150g) canned chickpeas, drained
10.8g



150g Tofu, stir-fried
26.9g



½ cup peas (85g), boiled
4.1g

Source: New Zealand FOODfiles™ 2018 Version 01[2]

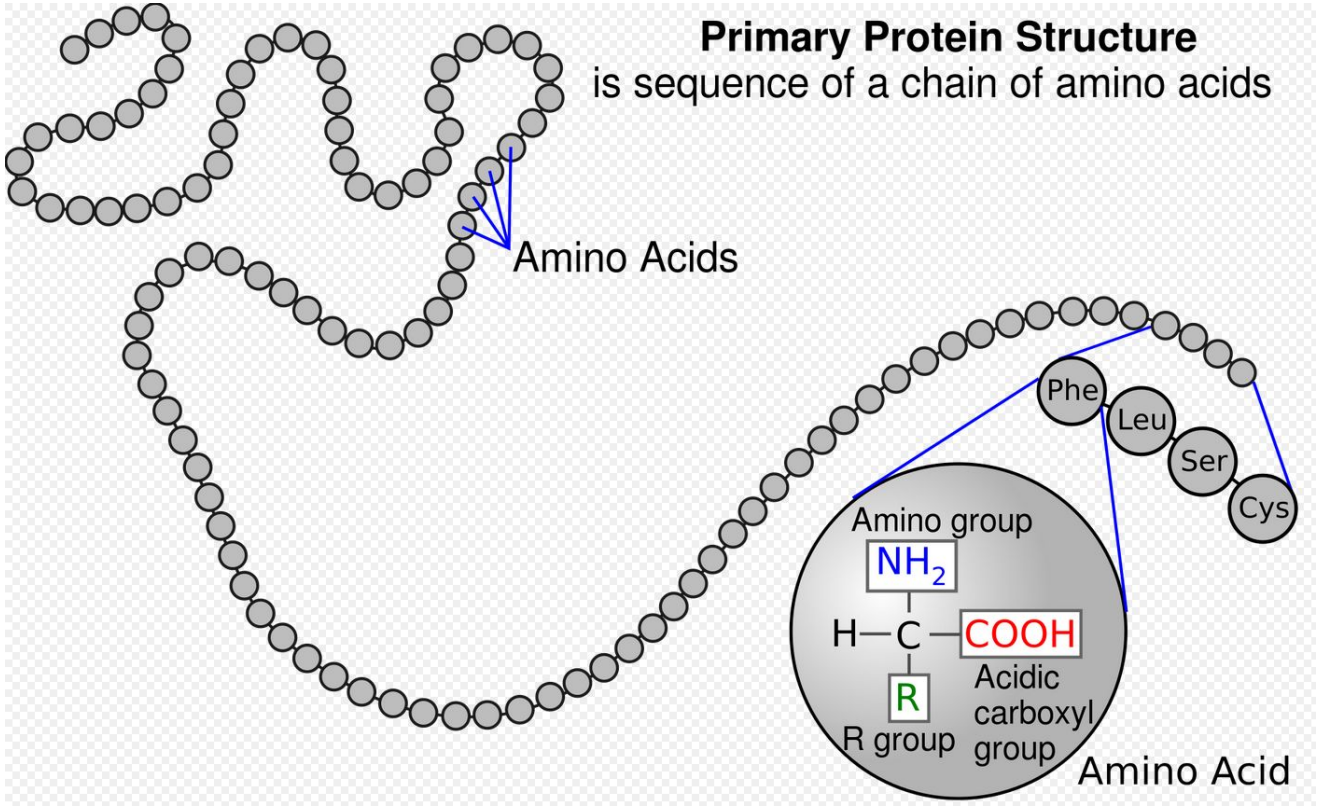
What are proteins?

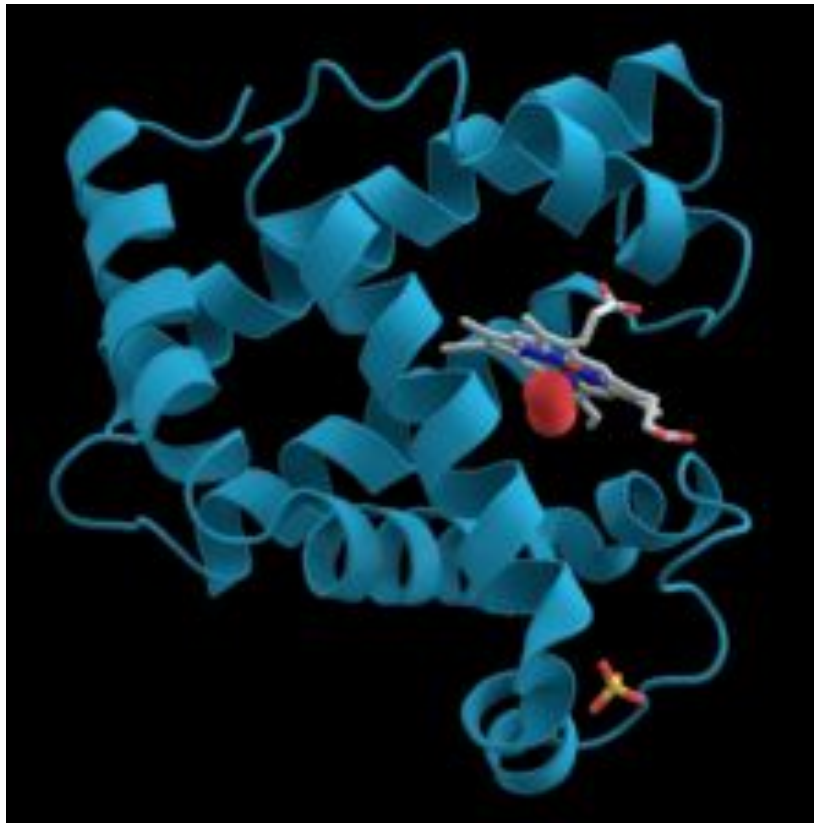
- Proteins are large biomolecules and macromolecules that comprise long chains of amino acids, linked by the peptide bond.

- Proteins perform a vast number of functions in our body such as catalysing metabolic reactions, DNA replication, providing structural support to the cells, transporting molecules.

- Proteins are nanomachines of the cells.

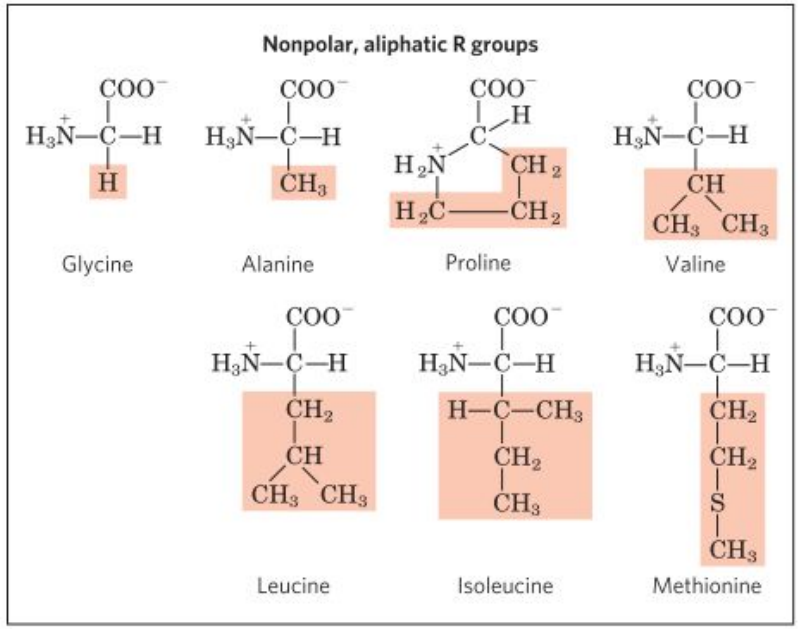
Motor protein walking on a microtubule:
https://www.youtube.com/watch?v=y-uuk4Pr2i8
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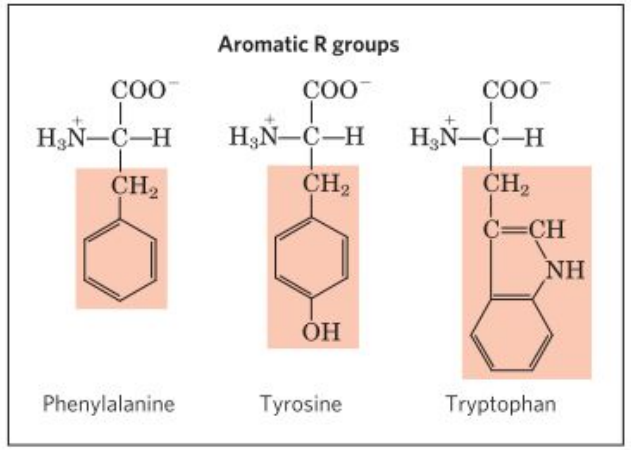


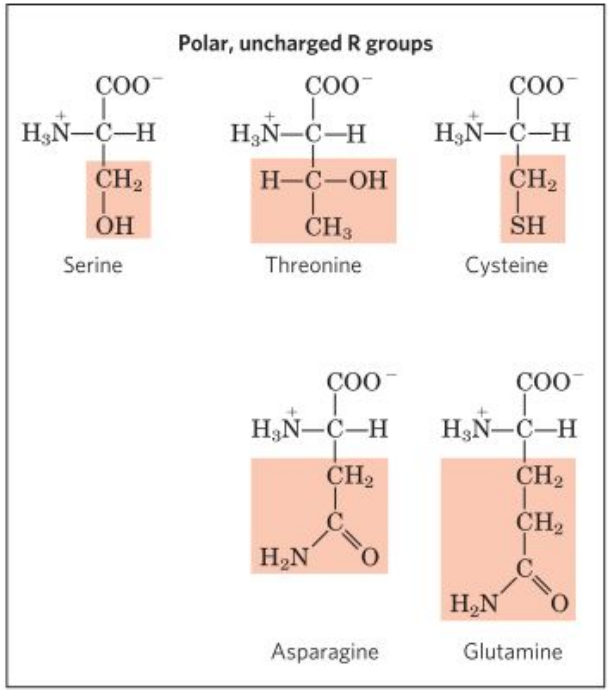


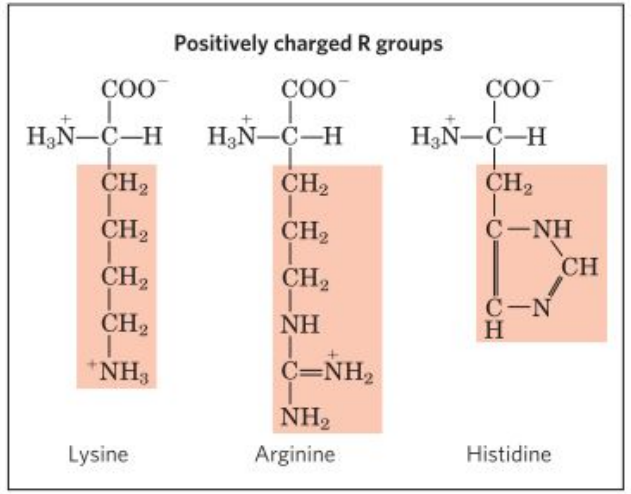
The 20 common amino acids of proteins

- Amino acids are organic compounds that contain amino (-NH3+) and carboxyl (-COO-) functional groups, along with a side chain (R group) specific to each amino acid.
- They are classified based on their polarity, ionization and side chain group type (aliphatic, acyclic, aromatic).
- Natural amino acids have the L configuration.
- There are several Uncommon amino acids such as 4-hydroxyproline, 5-hydroxylysin, 6-N-Methyllysine, Selenocystein. See page no. 82 in Lehninger.









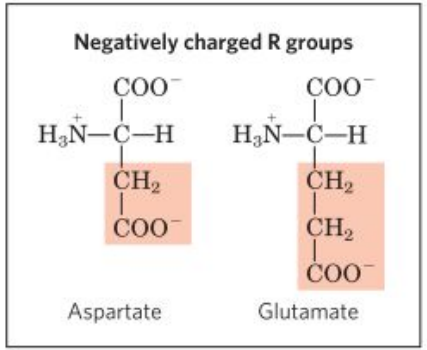
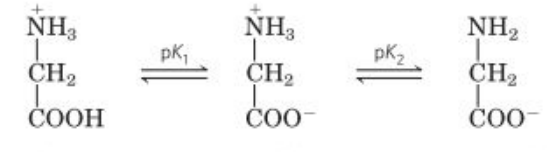


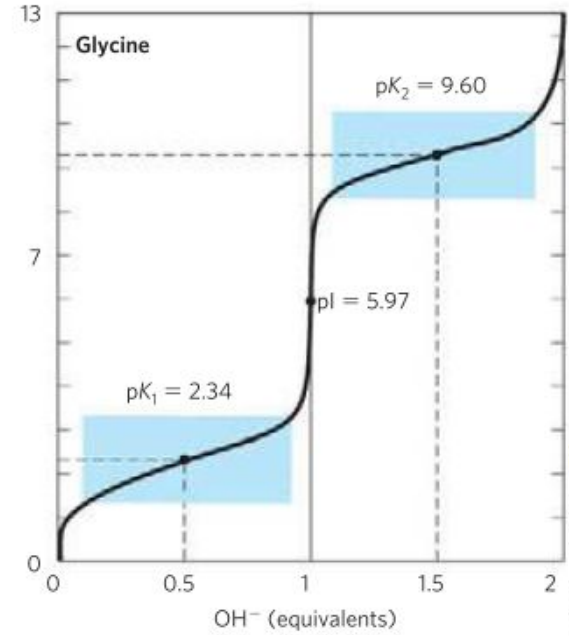
Table of standard amino acid abbreviations and properties

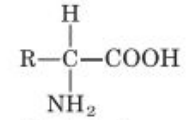
Amino acid	3- and 1-letter symbols		Side chain			Hydropathy index[34]	Molar absorptivity[35]		Molecular mass	Abundance in proteins (%)[36]	Standard genetic coding, IUPAC notation
	3	1	Class	Polarity[37]	Net charge at pH 7.4[37]		Wavelength, λ_{max} (nm)	Coefficient ε ($mM^{-1} \cdot cm^{-1}$)			
Alanine	Ala	A	Aliphatic	Nonpolar	Neutral	1.8			89.094	8.76	GCN
Arginine	Arg	R	Fixed cation	Basic polar	Positive	−4.5			174.203	5.78	MGR, CGY[38]
Asparagine	Asn	N	Amide	Polar	Neutral	−3.5			132.119	3.93	AAY
Aspartate	Asp	D	Anion	Brønsted base	Negative	−3.5			133.104	5.49	GAY
Cysteine	Cys	C	Thiol	Brønsted acid	Neutral	2.5	250	0.3	121.154	1.38	UGY
Glutamine	Gln	Q	Amide	Polar	Neutral	−3.5			146.146	3.9	CAR
Glutamate	Glu	E	Anion	Brønsted base	Negative	−3.5			147.131	6.32	GAR
Glycine	Gly	G	Aliphatic	Nonpolar	Neutral	−0.4			75.067	7.03	GGN
Histidine	His	H	Aromatic cation	Brønsted acid and base	Positive, 10% Neutral, 90%	−3.2	211	5.9	155.156	2.26	CAY
Isoleucine	Ile	I	Aliphatic	Nonpolar	Neutral	4.5			131.175	5.49	AUH
Leucine	Leu	L	Aliphatic	Nonpolar	Neutral	3.8			131.175	9.68	YUR, CUY[39]
Lysine	Lys	K	Cation	Brønsted acid	Positive	−3.9			146.189	5.19	AAR
Methionine	Met	M	Thioether	Nonpolar	Neutral	1.9			149.208	2.32	AUG
Phenylalanine	Phe	F	Aromatic	Nonpolar	Neutral	2.8	257, 206, 188	0.2, 9.3, 60.0	165.192	3.87	UUY
Proline	Pro	P	Cyclic	Nonpolar	Neutral	−1.6			115.132	5.02	CCN
Serine	Ser	S	Hydroxylic	Polar	Neutral	−0.8			105.093	7.14	UCN, AGY
Threonine	Thr	T	Hydroxylic	Polar	Neutral	−0.7			119.119	5.53	ACN
Tryptophan	Trp	W	Aromatic	Nonpolar	Neutral	−0.9	280, 219	5.6, 47.0	204.228	1.25	UGG
Tyrosine	Tyr	Y	Aromatic	Brønsted acid	Neutral	−1.3	274, 222, 193	1.4, 8.0, 48.0	181.191	2.91	UAY
Valine	Val	V	Aliphatic	Nonpolar	Neutral	4.2			117.148	6.73	GUN

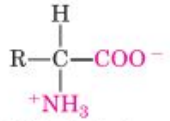
Amino Acids have characteristic titration curves

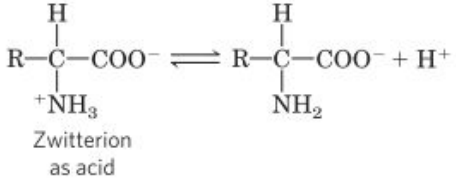
- Amino acids have two ionizable groups, the carboxyl group and the amino group.
- When you titrate with a strong base (NaOH), the plot shows two distinct stages, corresponding to deprotonation of two different groups of the amino acids.
- The pKa is a measure of the tendency of a group to give up a proton, with that tendency decreases tenfold as the pKa increases by one unit.
- For Glycine, p*K*a of COOH group is 2.34, whereas p*K*a of NH2 group is 9.6
- The carboxyl group of glycine is over 100 times more acidic (more easily ionized) than the carboxyl group of acetic acid (p*K*a = 4.76).
- The titration curve of glycine indicates that this amino acid has two regions of buffering power. One of these is the relatively flat portion of the curve, 1 pH unit on either side of the pK_1 of 2.34 and the second one is near pK_2 of 9.60.

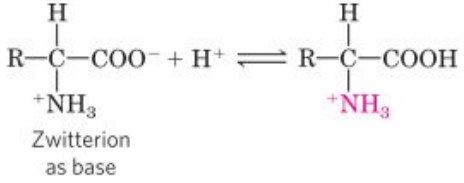












Titration curve predict the electric charge of amino acids

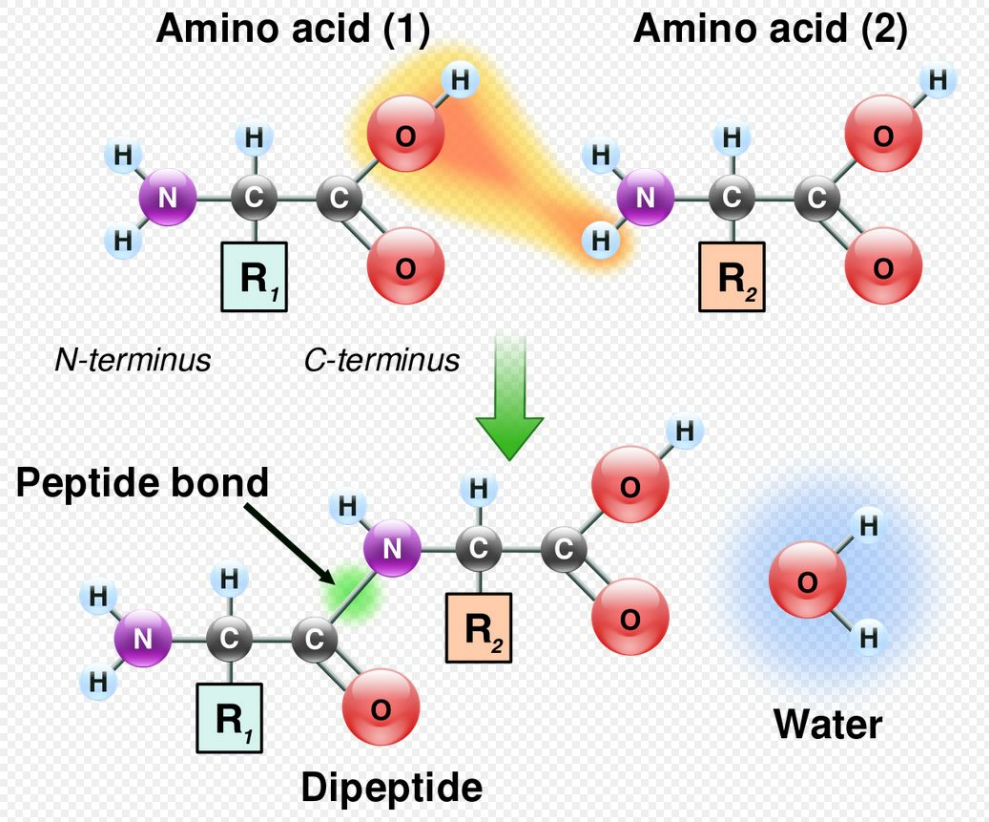
- At pH 5.97, glycine is present predominantly as its dipolar form, fully ionized but with no net electrical charge.

- The characteristic pH at which the net electric charge is zero is called the isoelectric point or isoelectric pH, designated pI. At this pH, the amino acid exists in its zwitterionic form.

$$\mathrm{pI} = \frac{1}{2}(\mathrm{p}K_1 + \mathrm{p}K_2) = \frac{1}{2}(2.34 + 9.60) = 5.97$$

- However, amino acids differ in their acid-base properties, based on the ionization capacity of the R group. The titration curve of all the amino acids are not the same. See page no. 84-85 in Lehninger. Titration curves for glutamate and histidine.

- Amino acids are linked together by the peptide bond to form a peptide (two/three/four/five amino acids)/oligopeptide (a few amino acids)/polypeptide (many amino acids)/protein (thousands of amino acids).



The pentapeptide, Ser-Gly-Tyr-Ala-Leu, or SGYAL

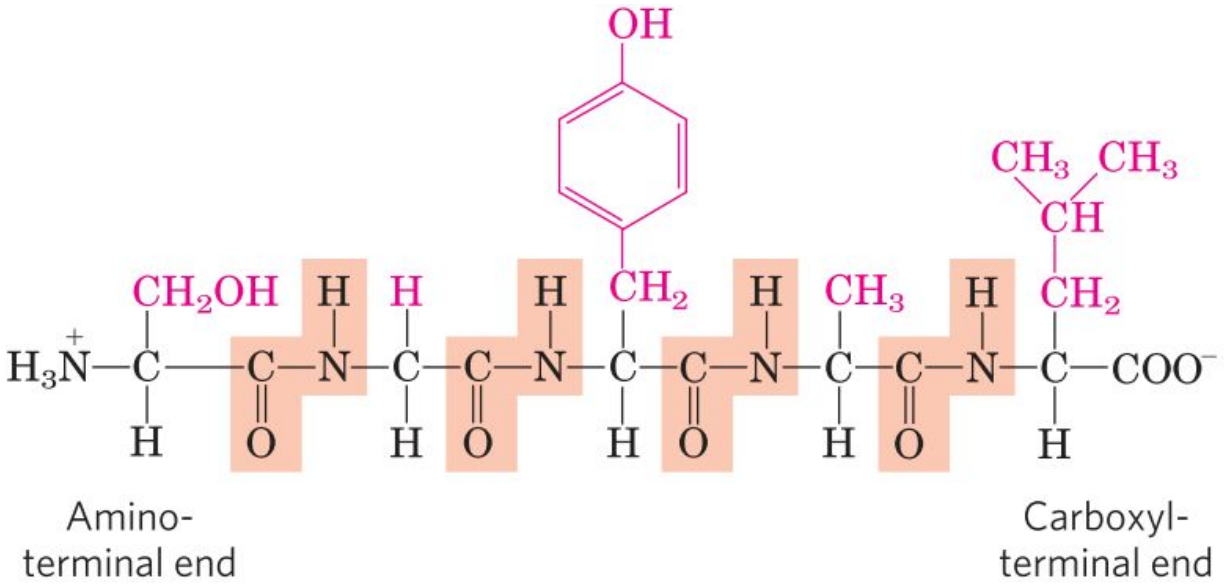


TABLE 3–2 Molecular Data on Some Proteins

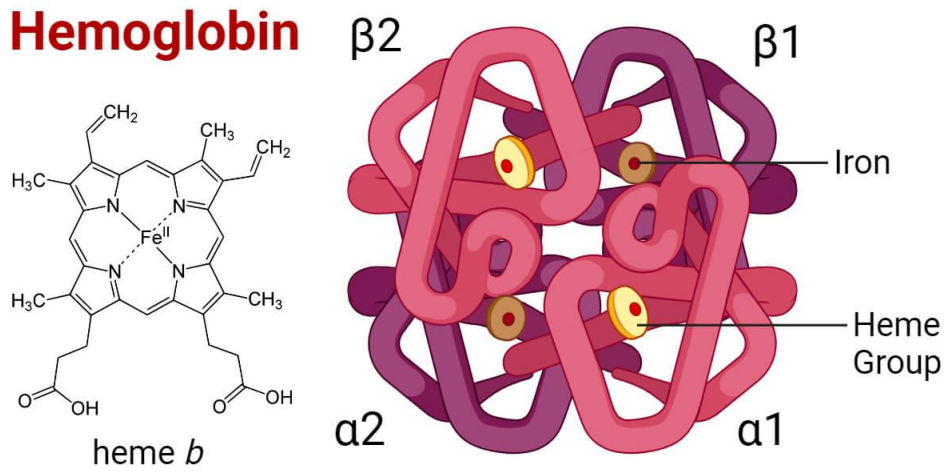
	Molecular weight	Number of residues	Number of polypeptide chains
Cytochrome *c* (human)	12,400	104	1
Ribonuclease A (bovine pancreas)	13,700	124	1
Lysozyme (chicken egg white)	14,300	129	1
Myoglobin (equine heart)	16,700	153	1
Chymotrypsin (bovine pancreas)	25,200	241	3
Chymotrypsinogen (bovine)	25,700	245	1
Hemoglobin (human)	64,500	574	4
Serum albumin (human)	66,000	609	1
Hexokinase (yeast)	107,900	972	2
RNA polymerase (*E. coli*)	450,000	4,158	5
Apolipoprotein B (human)	513,000	4,536	1
Glutamine synthetase (*E. coli*)	619,000	5,628	12
Titin (human)	2,993,000	26,926	1

Prosthetic group

- Some proteins contain chemical groups other than amino acids. This chemical group is known as prosthetic group.

TABLE 3–4 Conjugated Proteins

Class	Prosthetic group	Example
Lipoproteins	Lipids	β_1-Lipoprotein of blood
Glycoproteins	Carbohydrates	Immunoglobulin G
Phosphoproteins	Phosphate groups	Casein of milk
Hemoproteins	Heme (iron porphyrin)	Hemoglobin
Flavoproteins	Flavin nucleotides	Succinate dehydrogenase
Metalloproteins	Iron	Ferritin
	Zinc	Alcohol dehydrogenase
	Calcium	Calmodulin
	Molybdenum	Dinitrogenase
	Copper	Plastocyanin



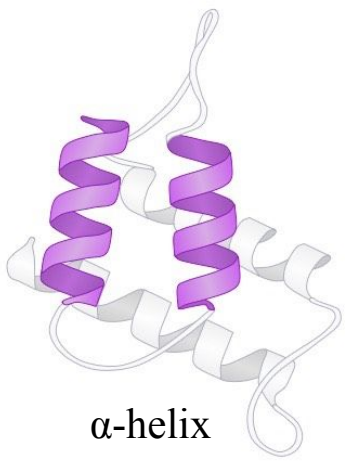
The structure of proteins

Primary structure: the linear sequence of amino acids: Gly-Ala-Lys-Trp

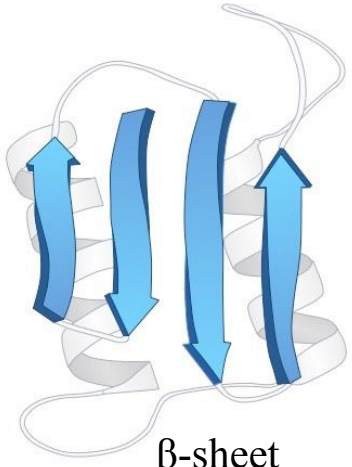
Secondary structure: stable arrangement of amino acids giving rise to recurring structural patterns, such as α-helix and β-sheet.

Tertiary structure: three-dimensional folding of a polypeptide chain.

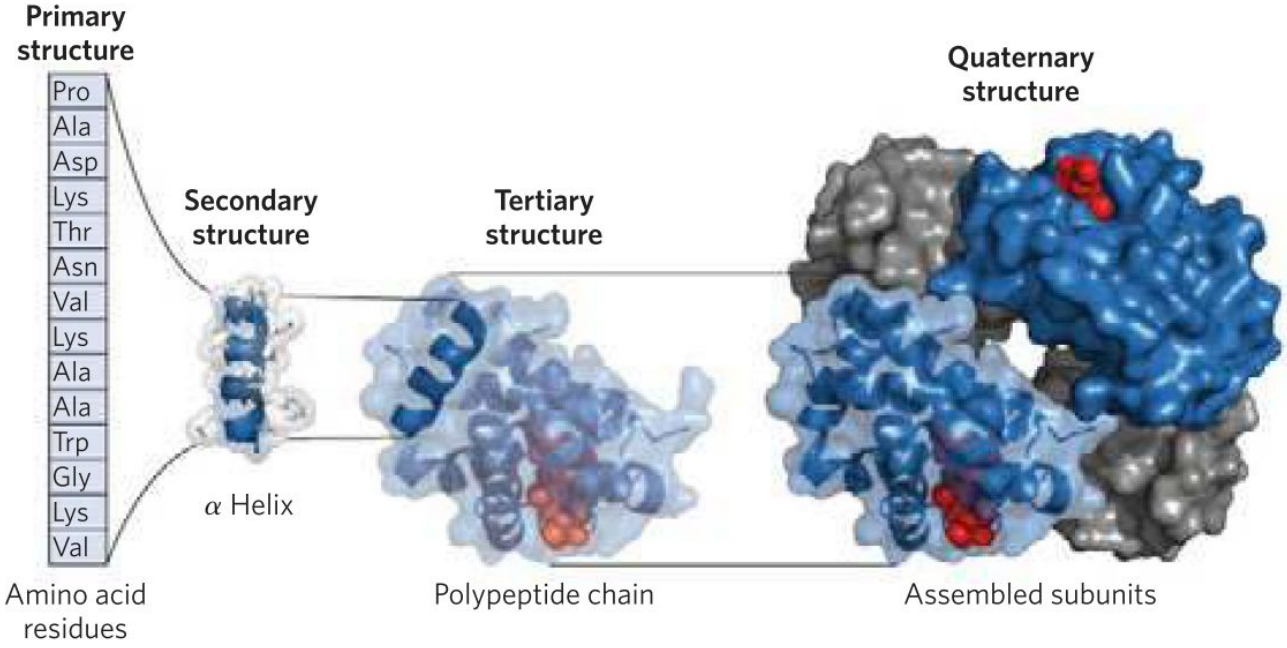
Quaternary structure: when a protein has two or more polypeptide chains, their arrangement in space.



α-helix



β-sheet



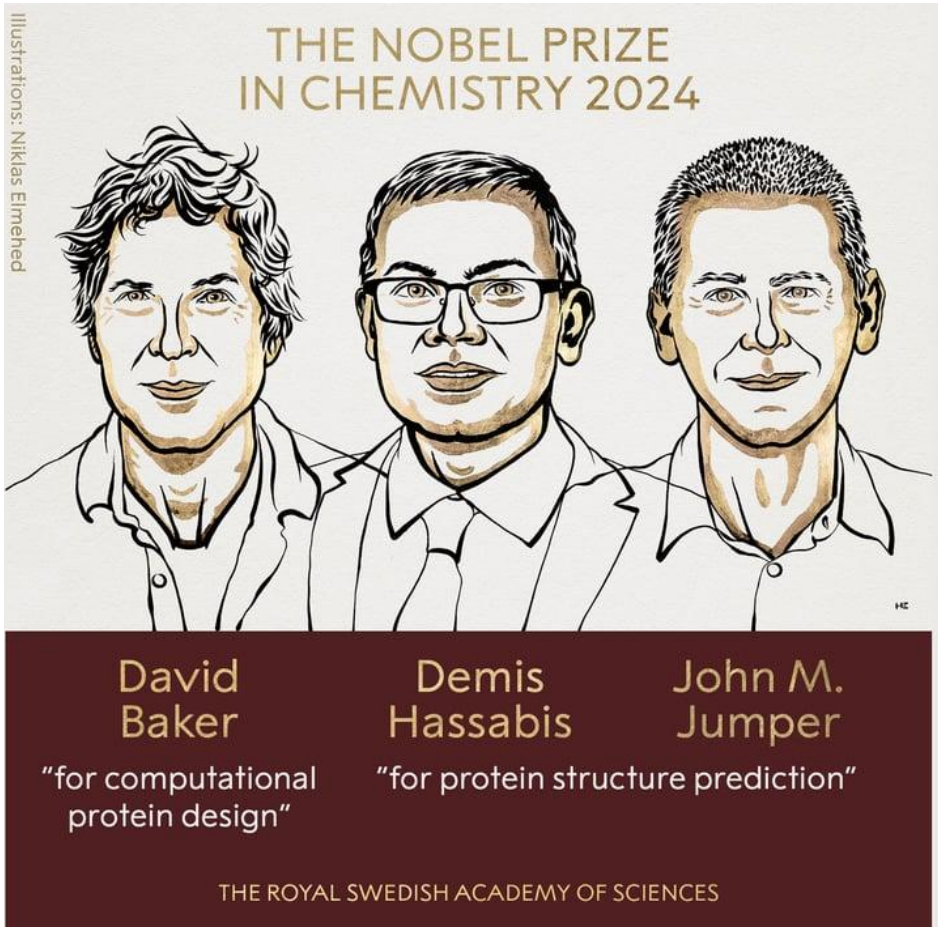
Nobel prize 2024 in Chemistry for AI-based protein structure prediction tool (Alpha Fold)

Demis Hassabis, John Jumper and David Baker for being awarded the 2024 Nobel Prize in Chemistry for their work on protein structure prediction and computational protein design.

Artificial intelligence pioneer Demis Hassabis, whose AI software cracked an impossible chemistry problem—predicting the structure of every protein known to science.

John Michael Jumper (born 1985) is an American chemist and computer scientist. He currently serves as director at Google DeepMind.[2][3][4] Jumper and his colleagues created AlphaFold,[5] an artificial intelligence (AI) model to predict protein structures from their amino acid sequence with high accuracy.

David Baker is an American biochemist and computational biologist who has pioneered methods to design proteins and predict their three-dimensional structures.



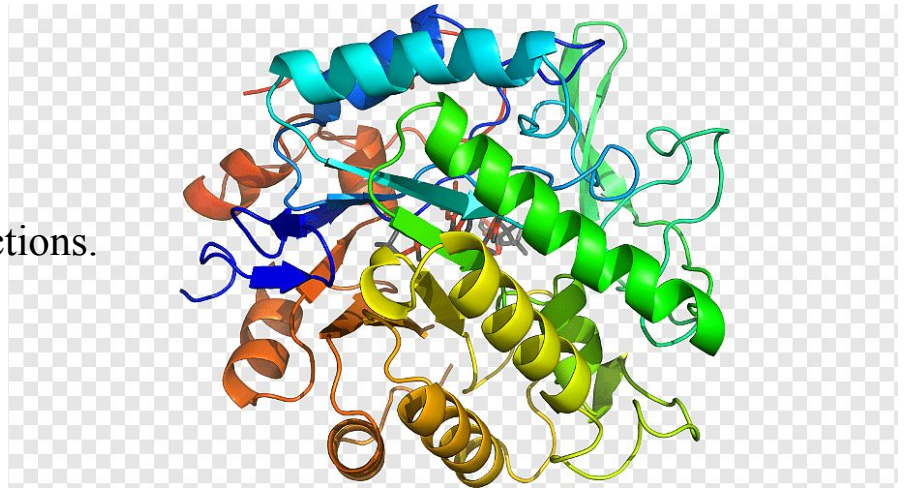
Our Hairs and Nails are proteins

- Our hairs and nails are made up of a protein, known as α-Keratin. Hair grows at a rate of 15-20 cm/year, whereas nails grow 3.5 mm each month.

- The fundamental structural element of α-Keratin is the α-helix.

What are Enzymes?

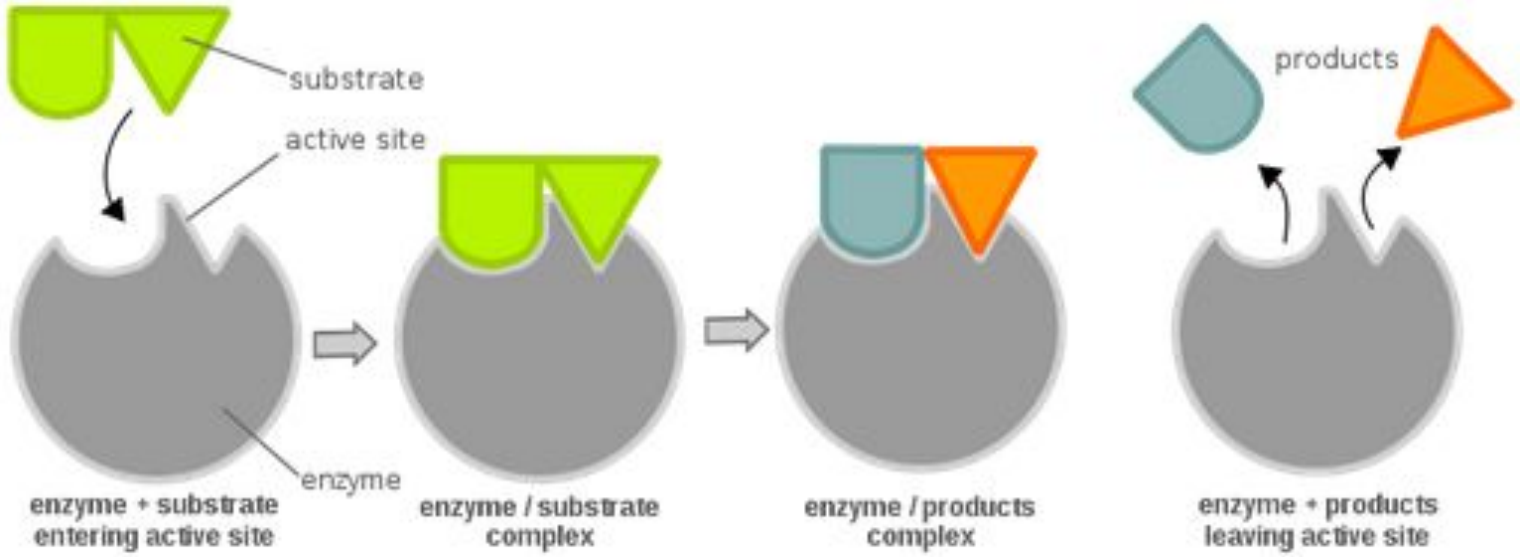
- Enzymes are proteins.
- Enzymes are bio-catalysts.
- Catalysts accelerate chemical reactions. Similarly, enzymes accelerate biochemical reactions.
- Like all catalysts, enzymes increase the reaction rate by lowering its activation energy.



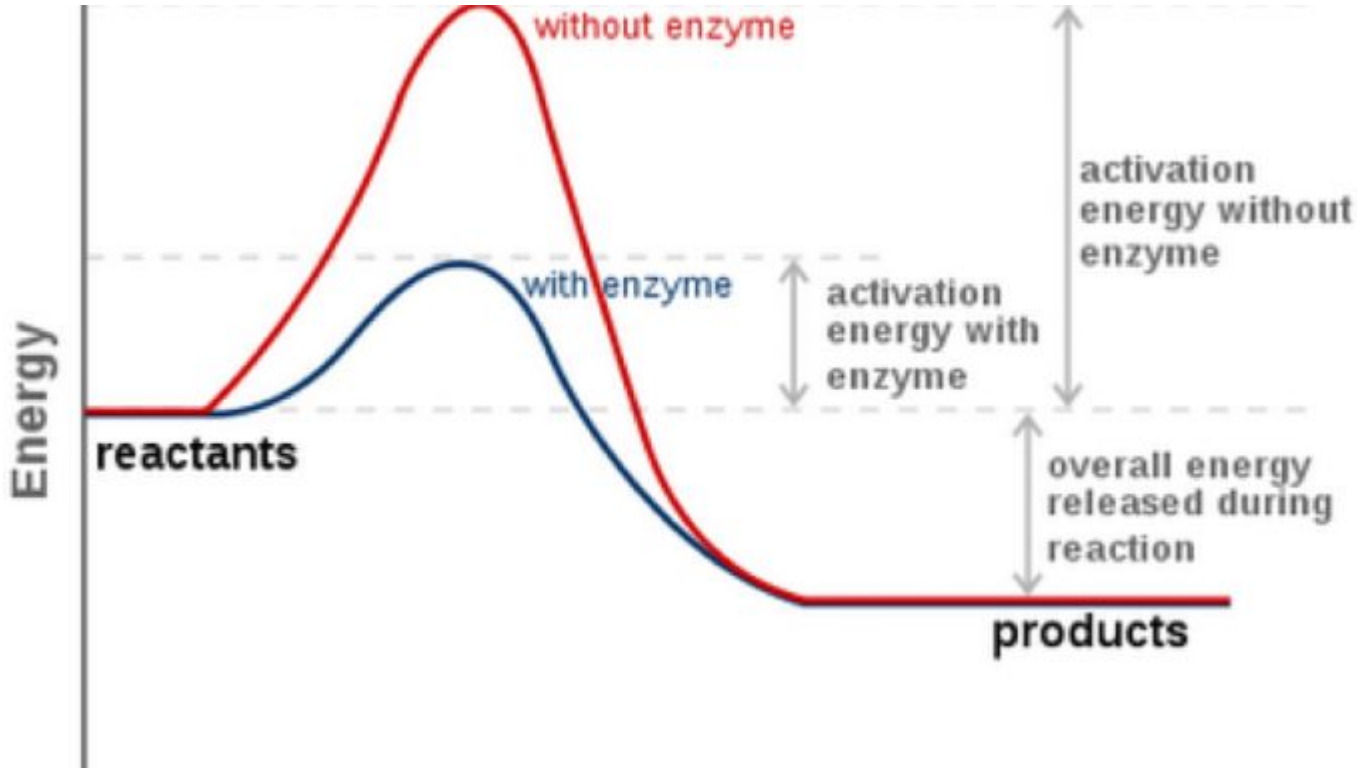
Enzymes Affect Reaction Rates, Not Equilibria

A simple enzymatic reaction might be written

$$E + S \rightleftharpoons ES \rightleftharpoons EP \rightleftharpoons E + P$$



Activation Energy



Activation Energy: the difference between the energy levels of the ground state and the transition state is known as the activation energy (ΔG).

Enzymes enhance the reaction rate by lowering activation energy.

International Classification of Enzymes

Enzymes are classified according to the type of reaction they catalyze.

TABLE 6–3 International Classification of Enzymes

Class no.	Class name	Type of reaction catalyzed
1	Oxidoreductases	Transfer of electrons (hydride ions or H atoms)
2	Transferases	Group transfer reactions
3	Hydrolases	Hydrolysis reactions (transfer of functional groups to water)
4	Lyases	Cleavage of C—C, C—O, C—N, or other bonds by elimination, leaving double bonds or rings, or addition of groups to double bonds
5	Isomerases	Transfer of groups within molecules to yield isomeric forms
6	Ligases	Formation of C—C, C—S, C—O, and C—N bonds by condensation reactions coupled to cleavage of ATP or similar cofactor

Enzyme Kinetics

Substrate concentration [S] affects the rate of enzyme-catalyzed reactions.

However, the substrate concentration [S] keeps changing in the enzymatic reaction as products are formed.

So, the initial velocity (Vo) is measured.

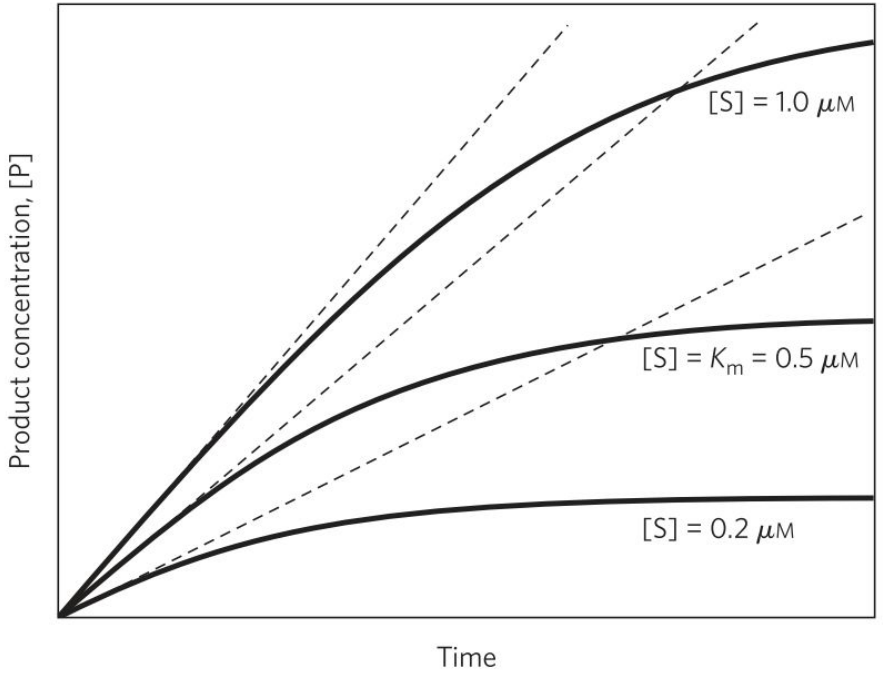


FIGURE 6–10 Initial velocities of enzyme-catalyzed reactions.

Kinetic parameters for an enzymatic reaction

The effect on V_0 of varying [S] when the enzyme concentration is held constant.

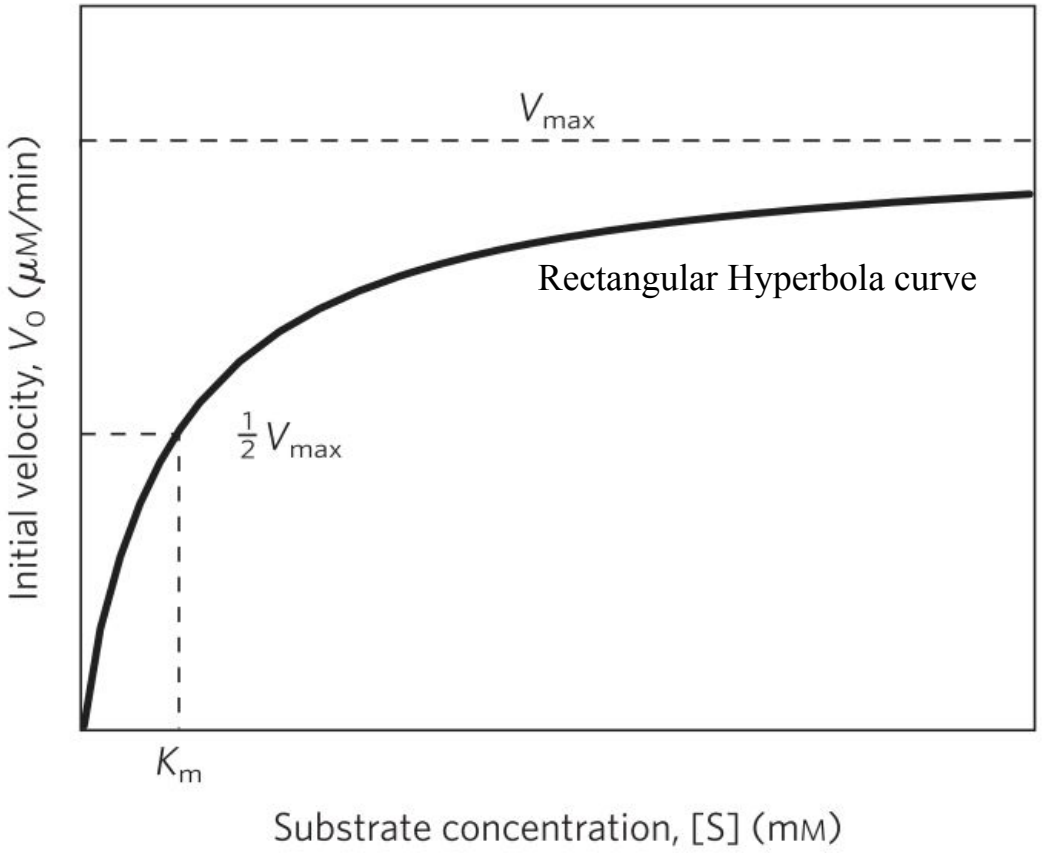


TABLE 6–6 K_m for Some Enzymes and Substrates

Enzyme	Substrate	K_m (mM)
Hexokinase (brain)	ATP	0.4
	D-Glucose	0.05
	D-Fructose	1.5
Carbonic anhydrase	HCO_3^-	26
Chymotrypsin	Glycyltyrosinylglycine	108
	N-Benzoyltyrosinamide	2.5
β-Galactosidase	D-Lactose	4.0
Threonine dehydratase	L-Threonine	5.0

TABLE 6–7 Turnover Number, k_{cat}, of Some Enzymes

Enzyme	Substrate	k_{cat} (s^{-1})
Catalase	H_2O_2	40,000,000
Carbonic anhydrase	HCO_3^-	400,000
Acetylcholinesterase	Acetylcholine	14,000
β-Lactamase	Benzylpenicillin	2,000
Fumarase	Fumarate	800
RecA protein (an ATPase)	ATP	0.5

V_{max} = maximum velocity (unit μM/min)

K_m = The substrate concentration at which the velocity is Vmax/2. (unit mM) $K_m = [S]$ when $V_0 = \frac{1}{2}V_{max}$

K_{cat} = turnover number (the number of substrate molecules converted to product in a given unit of time on a single enzyme molecule when the enzyme is saturated with substrate. (unit s^{-1})

The relationship between Substrate Concentration [S] and Reaction Rate (V_0)

The rectangular hyperbola curve can be expressed algebraically by the Michaelis-Menten Equation.

$$V_0 = \frac{V_{max}[S]}{K_m + [S]}$$

where, K_m is the Michaelis constant.

The value of KM is inversely related to the affinity of the enzyme for its substrate. **High values of KM correspond to low enzyme affinity for substrate** (it takes more substrate to get to Vmax).

TABLE 6–6 K_m for Some Enzymes and Substrates

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	N-Benzoyltyrosinamide	2.5
β-Galactosidase	D-Lactose	4.0
Threonine dehydratase	L-Threonine	5.0

Lineweaver-Burk equation/plot for calculating Km and Vmax

Michaelis-Menten equation

$$V_0 = \frac{V_{max}[S]}{K_m + [S]}$$

Transformed

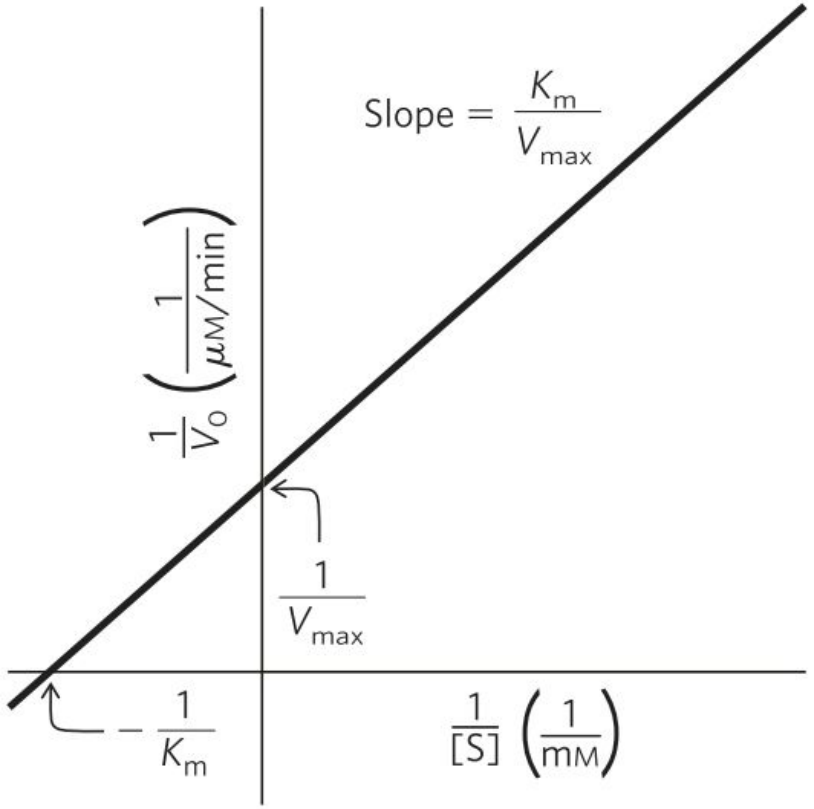
Lineweaver-Burk equation

$$\frac{1}{V_0} = \frac{K_m}{V_{max}[S]} + \frac{1}{V_{max}}$$

Plot of $1/V_0$ versus $1/[S]$ (double reciprocal) yields a straight line.

This line has a slope of K_m/V_{max}

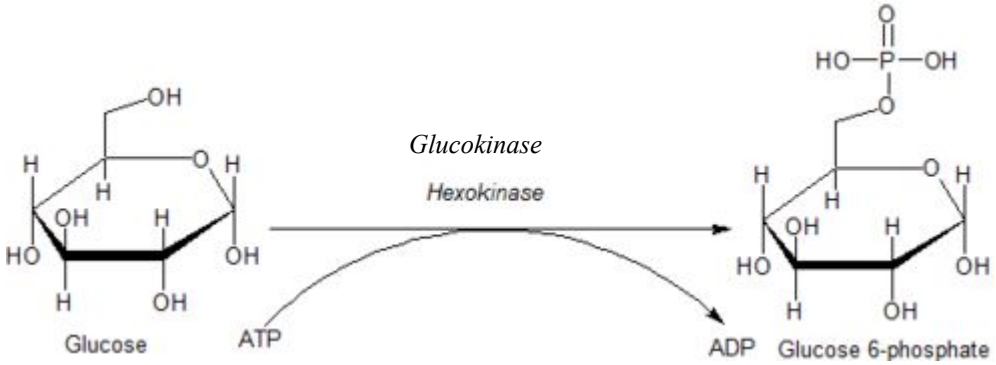
Accurate determination of V_{max}

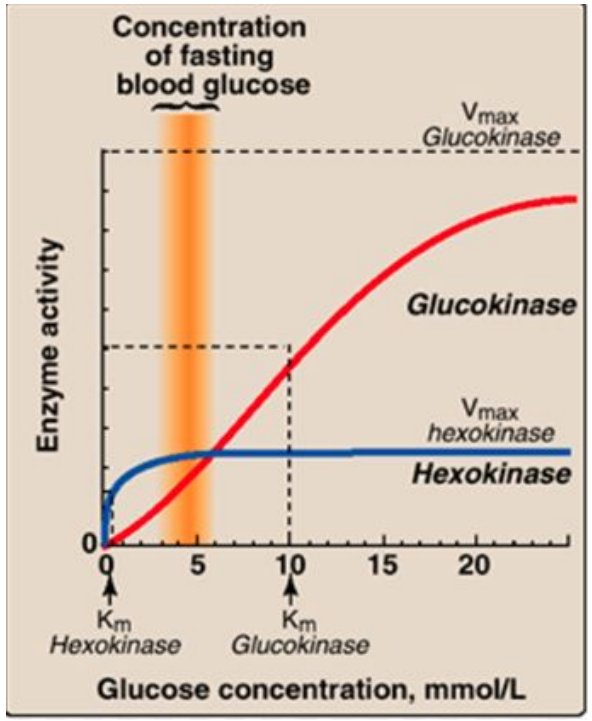


A double-reciprocal or Lineweaver-Burk plot.

Enzymes with same function but different kinetics

Hexokinase and Glucokinase, both are enzymes that phosphorylates glucose.





Difference between Hexokinase and Glucokinase

	Hexokinase	Glucokinase
Substrate specificity	All hexoses	Mainly Glucose
Km	Low (high affinity) *Works at normal glucose concentration*	High (low affinity) *works only when glucose levels are elevated*
Location	Universal	Mainly liver and Beta cells of pancreas
Vmax (rate of reaction)	Low	High
Glucose-6-PO4 (Allosteric inhibition)	Inhibits the enzyme	No inhibition
Insulin	No regulation	Positive regulation

Comparing catalytic mechanisms and efficiencies of two different enzymes

- The kinetic parameters K_{cat} and K_m are useful for comparing different enzymes.
- The best way to compare the catalytic efficiencies of different enzymes or the turnover of different substrates by the same enzyme is to compare the ratio of K_{cat}/K_m for the two reactions.
- This ratio is known as Specificity Constant.
- The upper limit to K_{cat}/K_m, imposed by the rate at which E and S can diffuse together in an aqueous solution. This diffusion controlled limit is 10^8 to 10^9 $M^{-1}s^{-1}$.

TABLE 6–8 Enzymes for Which k_{cat}/K_m Is Close to the Diffusion-Controlled Limit (10^8 to 10^9 $M^{-1}s^{-1}$)

Enzyme	Substrate	K_{cat} (s^{-1})	K_m (M)	K_{cat}/K_m ($M^{-1}s^{-1}$)
Acetylcholinesterase	Acetylcholine	1.4×10^4	9×10^{-5}	1.6×10^8
Carbonic anhydrase	CO_2	1×10^6	1.2×10^{-2}	8.3×10^7
	HCO_3^-	4×10^5	2.6×10^{-2}	1.5×10^7
Catalase	H_2O_2	4×10^7	1.1×10^0	4×10^7
Crotonase	Crotonyl-CoA	5.7×10^3	2×10^{-5}	2.8×10^8
Fumarase	Fumarate	8×10^2	5×10^{-6}	1.6×10^8
	Malate	9×10^2	2.5×10^{-5}	3.6×10^7
β-Lactamase	Benzylpenicillin	2.0×10^3	2×10^{-5}	1×10^8

Catalytic perfection

WORKED EXAMPLE 6–1 Determination of K_m

An enzyme is discovered that catalyzes the chemical reaction

$$\text{SAD} \rightleftharpoons \text{HAPPY}$$

A team of motivated researchers sets out to study the enzyme, which they call happyase. They find that the k_{cat} for happyase is 600 s^{-1}. They carry out several experiments.

When $[E_t]$ = 20 nM and [SAD] = 40 μM, the reaction velocity, V_0, is 9.6 $\mu M\ s^{-1}$. Calculate K_m for the substrate SAD.

Solution: We know k_{cat}, $[E_t]$, [S], and V_0. We want to solve for K_m. Equation 6–26, in which we substitute $k_{cat}[E_t]$ for V_{max} in the Michaelis-Menten equation, is most useful here. Substituting our known values in Equation 6–26 allows us to solve for K_m.

$$V_0 = \frac{k_{cat}[E_t][S]}{K_m + [S]}$$

$$9.6\ \mu M\ s^{-1} = \frac{(600\ s^{-1})(0.020\ \mu M)(40\ \mu M)}{K_m + 40\ \mu M}$$

$$9.6\ \mu M\ s^{-1} = \frac{480\ \mu M^2\ s^{-1}}{K_m + 40\ \mu M}$$

$$9.6\ \mu M\ s^{-1}(K_m + 40\ \mu M) = 480\ \mu M^2\ s^{-1}$$

$$K_m + 40\ \mu M = \frac{480\ \mu M^2\ s^{-1}}{9.6\ \mu M\ s^{-1}}$$

$$K_m + 40\ \mu M = 50\ \mu M$$

$$K_m = 50\ \mu M - 40\ \mu M$$

$$K_m = 10\ \mu M$$

Further reading

Book: Lehninger: Principles of Biochemistry, 6th edition. ISBN: 1464109621.

Chapter 3: Amino Acids, Peptides, and Proteins, Page 75-114.
Chapter 4: Three dimensional structure of proteins, Page
Chapter 6: Enzymes

Next class on 30/10/2025 (Thursday)